Network Packet Capture and Analysis Using Wireshark

Objective
The objective of this project is to capture and analyze network packets to identify protocols, understand TCP communication, and investigate potentially suspicious or malformed traffic.

Tools Used
- Wireshark network protocol analyzer
- Windows operating system
- Internet connection

Methodology
Network traffic was captured using Wireshark on an active network interface. Display filters were used to examine DNS queries, HTTP traffic, and TCP connections. Packet details, including source and destination IP addresses, port numbers, packet lengths, and TCP flags, were inspected.

Analysis
DNS packets were examined to understand domain name resolution. TCP packets were analyzed to identify the SYN, SYN-ACK, and ACK sequence used during connection establishment. HTTP packets were examined where available.

Potentially unusual activity was investigated by checking TCP resets, retransmissions, unexpected ports, repeated connection attempts, and malformed-packet warnings. Such observations require further investigation before being classified as malicious.

Findings
The actual findings should be completed after examining the captured traffic:
1. Protocols observed: [Enter protocols found].
2. TCP handshake: [Successful, incomplete, or not observed].
3. DNS activity: [Summarize queries and responses].
4. Unusual packets: [Describe observed anomalies or state that none were identified].
5. Possible indicators of compromise: [List evidence, if any].

Conclusion
Wireshark provides a useful method for understanding network communication and investigating unusual traffic. The analysis helps identify protocol behavior and potential network issues. However, suspicious packet patterns alone do not prove that a security incident has occurred.